EXPERIMENT 13

CREATING AND CONFIGURING A VIRTUAL MACHINE

Aim

To create a virtual machine using VMware Workstation and configure it with one CPU core, 2 GB RAM, and a 15 GB virtual hard disk.

Software Requirements

1. Windows host computer
2. VMware Workstation Pro
3. Ubuntu 18.04.3 LTS installation image
4. Internet connection

Procedure

1. Open VMware Workstation and select Create a New Virtual Machine.
2. Select Ubuntu 64-bit as the guest operating system.
3. Set the virtual hard disk capacity to 15 GB.
4. In Customize Hardware, set memory to 2048 MB and configure one processor with one core.
5. Finish creating the virtual machine and allow Ubuntu to install.
6. Log in to Ubuntu and verify the assigned resources using terminal commands.

Configuration and Verification

Component	Required Setting	Verified Result
CPU	1 CPU core	1 (nproc output)
Memory	2 GB RAM	2048 MB configured; 1.9 GiB detected
Virtual disk	15 GB	15G (sda)
Guest OS	Ubuntu 64-bit	Ubuntu 18.04.3 LTS

Verification commands: nproc checks available CPU processing units; free -h displays memory; lsblk lists storage devices and sizes.

VMware Hardware Settings Screenshot

Figure 1 shows the configured virtual machine hardware: 2048 MB memory, one processor, and a 15 GB virtual hard disk.

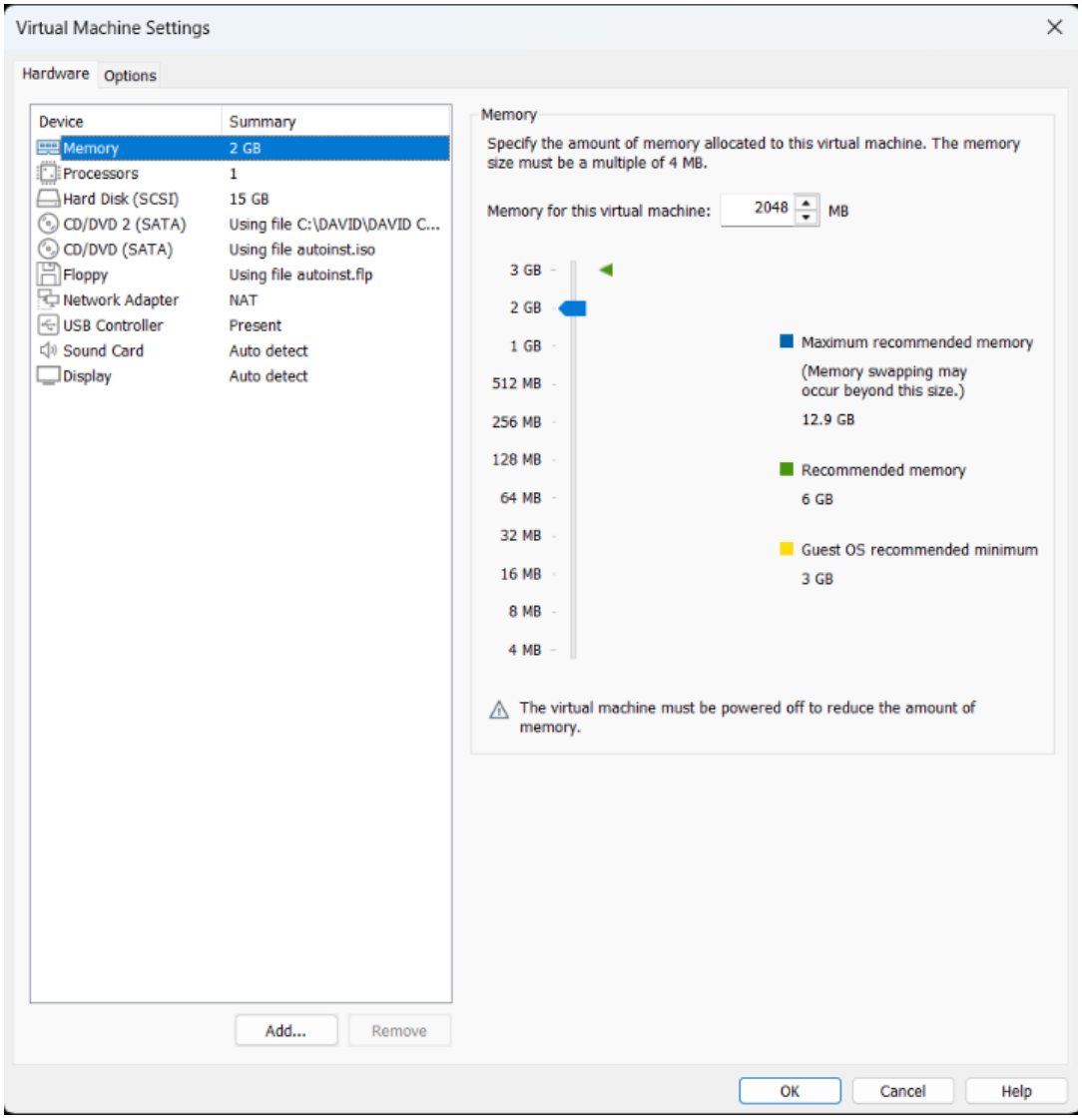


Figure 1: VMware settings showing 2 GB memory, one processor, and a 15 GB virtual disk.

Ubuntu Terminal Verification Screenshot

Figure 2 records the Ubuntu terminal commands used to verify CPU, memory, and disk capacity.



Figure 2: Ubuntu terminal showing nproc, free -h, and lsblk output.

Result

The virtual machine was created in VMware Workstation. The hardware settings confirm 2048 MB memory, one processor, and a 15 GB virtual hard disk. Terminal checks showed one available CPU processing unit, approximately 1.9 GiB of memory, and a 15 GB disk (sda). The required virtual machine configuration was verified.